Master’s degree in Control System Engineering

Reinforcement Learning
LAB 2

Dynamic Programming
&
car rental problem

DIPARTIMENTO
DI INGEGNERIA
DELL’INFORMAZIONE

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Policy Iteration (using iterative policy evaluation) for estimating $\pi \approx \pi_*$

1. Initialization
   $V(s) \in \mathbb{R}$ and $\pi(s) \in \mathcal{A}(s)$ arbitrarily for all $s \in \mathcal{S}$

2. Policy Evaluation
   Loop:
   - $\Delta \leftarrow 0$
   - Loop for each $s \in \mathcal{S}$:
     - $v \leftarrow V(s)$
     - $V(s) \leftarrow \sum_{s',r} p(s', r \,|\, s, \pi(s))\big[r + \gamma V(s')\big]$
     - $\Delta \leftarrow \max(\Delta, |v - V(s)|)$

   until $\Delta < \theta$ (a small positive number determining the accuracy of estimation)

3. Policy Improvement
   policy-stable $\leftarrow$ *true*
   For each $s \in \mathcal{S}$:
   - *old-action* $\leftarrow \pi(s)$
   - $\pi(s) \leftarrow \arg\max_a \sum_{s',r} p(s', r \,|\, s, a)\big[r + \gamma V(s')\big]$
   - If *old-action* $\neq \pi(s)$, then *policy-stable* $\leftarrow$ *false*

   If *policy-stable*, then stop and return $V \approx v_*$ and $\pi \approx \pi_*$; else go to 2

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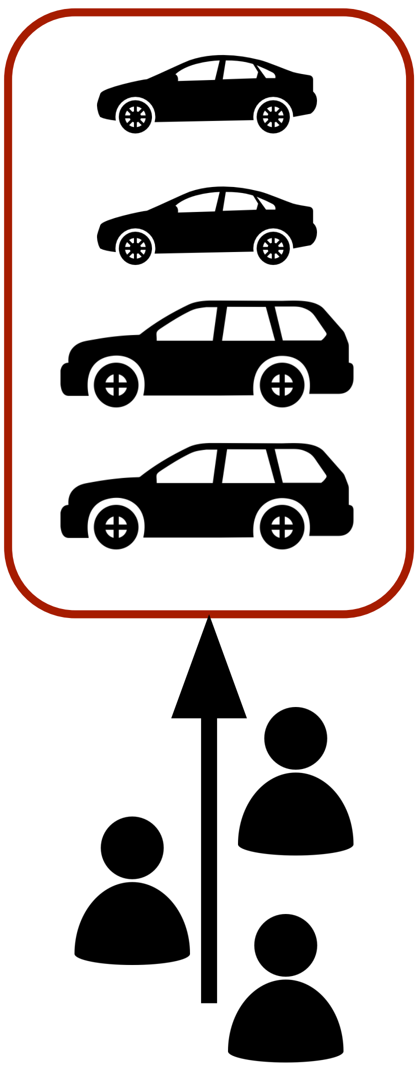
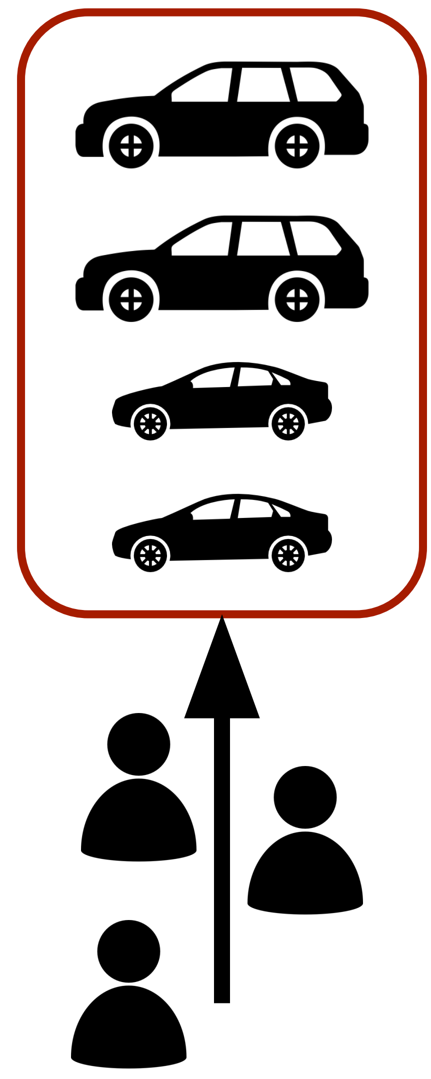
Jack's Car Rental

2 locations managed by Jack for a nationwide car rental company

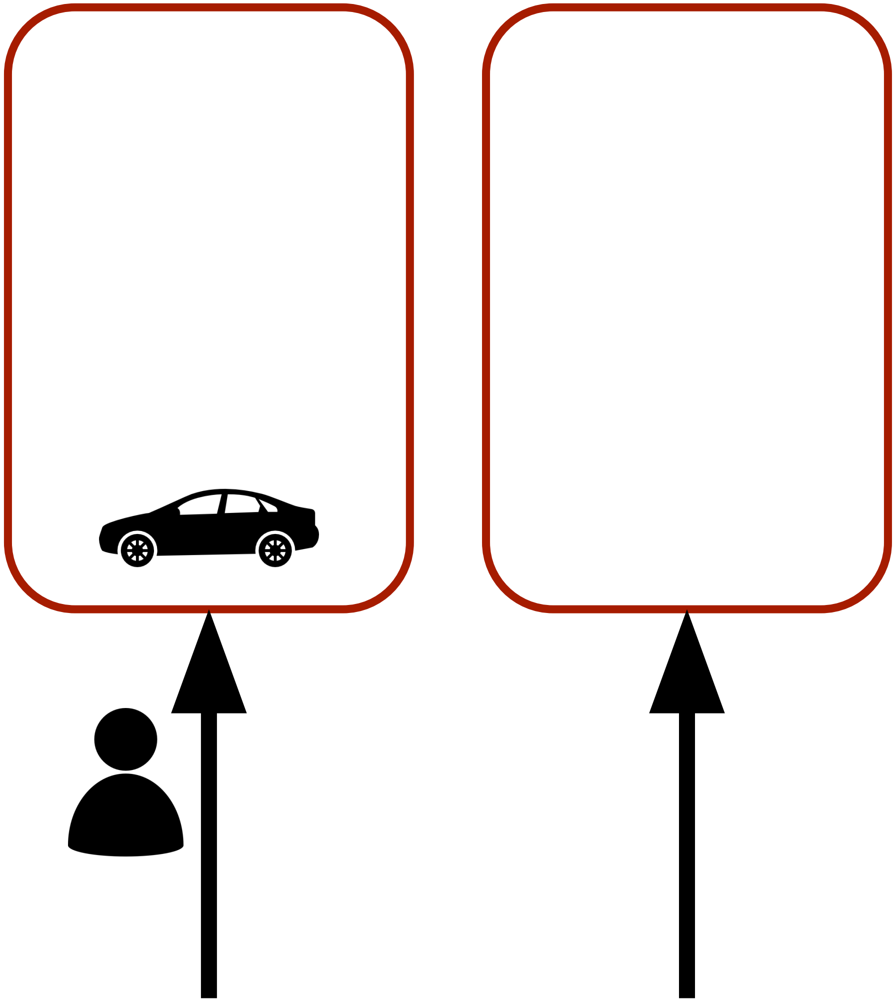


Customers arrive with an unknown distribution



Jack's Car Rental

If there is a car at one location a customer can arrive.



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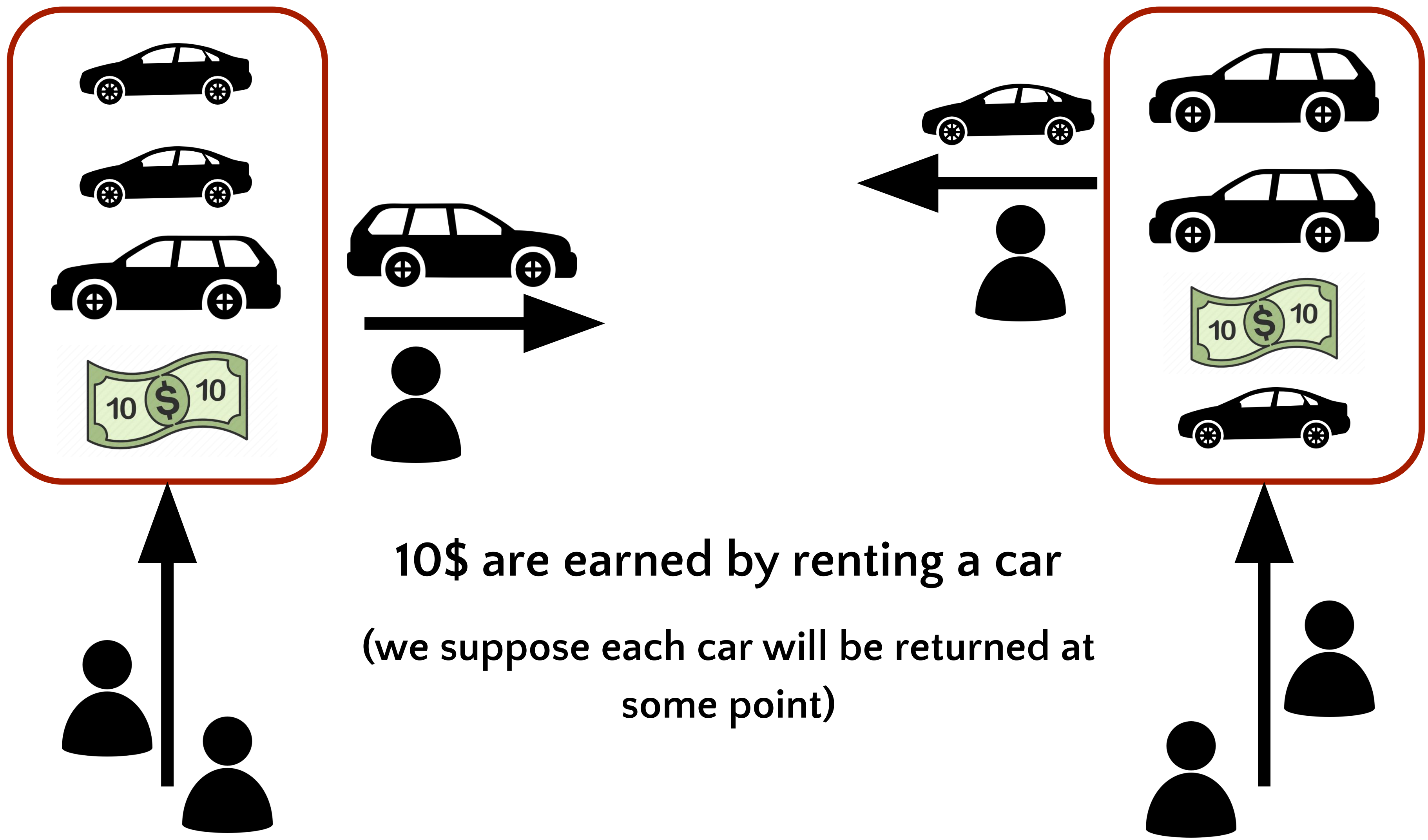

10$ are earned by renting a car

(we suppose each car will be returned at some point)

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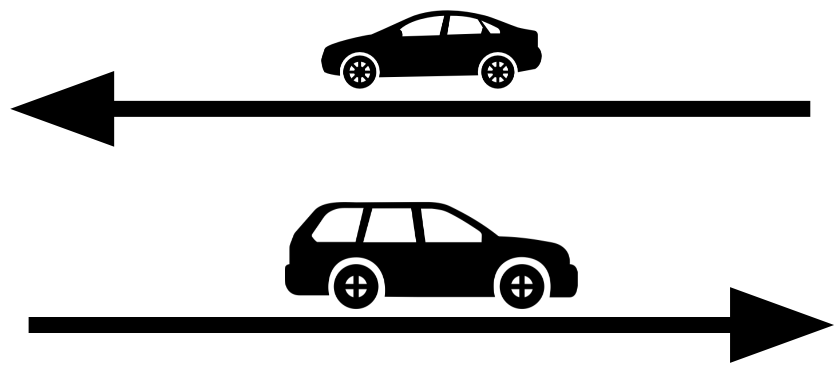


Jack can move the cars overnight, paying 2$ for each car

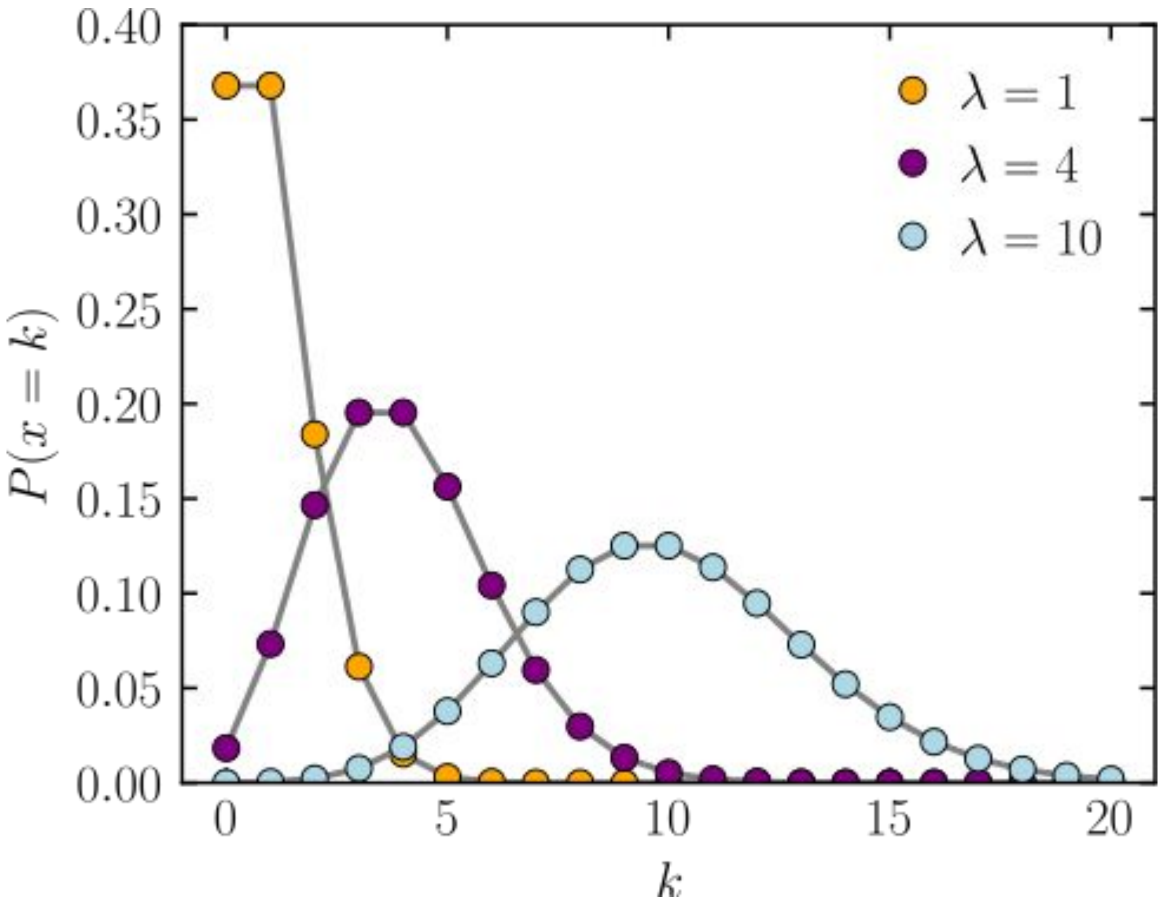
(we'll suppose there's an upper bound to the number of cars he can move)

Jack's Car Rental

Cars requested and returned are modelled either through a Poisson random variable (or through a constant value)

PMF = $\frac{\lambda^k e^{-\lambda}}{k!}$

(mean,variance=λ)



Jack's Car Rental

Jack can decide to move cars at the end of the working day!

$$a \in [-Max_M, Max_M]$$

```
actions = np.arange(-MAX_MOVE_OF_CARS, MAX_MOVE_OF_CARS + 1)
```

If a car is available and a customer arrives, he will rent the car!

(no questions asked)

Jack's Car Rental

The state of the environment is the pair of numer:

```
# [# of cars in first location, # of cars in second location]
```

$$s \in \mathbb{R}^2$$

The Value function of the environment is a Matrix!

$$V(s) \in Mat[\mathbb{R}]$$

Code for policy iteration algorithm

```
# Assumption
constant_returned_cars = True

# Initialization of the value-function
value = np.zeros((MAX_CARS + 1, MAX_CARS + 1))

# We start considering the simplest policy: for every possible state, no car is moved
# It makes sense: if is not clear what a meaningful action might be, better not to pay the
cost of moving cars!
policy = np.zeros(value.shape, dtype=np.int)

...
```

Code for policy iteration algorithm

```
...

while True:
    # policy evaluation (in-place)
    while True:
        old_value = value.copy()
        # Sweep through all states following the same policy
        for i in range(MAX_CARS + 1):
            for j in range(MAX_CARS + 1):
                new_state_value = expected_return([i, j], policy[i, j], value)
                # in-place update!
                value[i, j] = new_state_value
        max_value_change = abs(old_value - value).max()

        if max_value_change < eps_val:
            break

    ...
```

Code for policy iteration algorithm

```python
...

# policy improvement
policy_stable = True
for i in range(MAX_CARS + 1):
    for j in range(MAX_CARS + 1):
        old_action = policy[i, j]
        action_returns = []
        for action in actions:
            # if it is a 'legal' action,
            if (0 <= action <= i) or (-j <= action <= 0):
                action_returns.append(expected_return([i, j], action, value))
            else:
                action_returns.append(-np.inf)
        # Substitution with greedy policy at all states
        new_action = actions[np.argmax(action_returns)]
        policy[i, j] = new_action
        if policy_stable and old_action != new_action:
            policy_stable = False

# If stable instead
if policy_stable:
    break
```